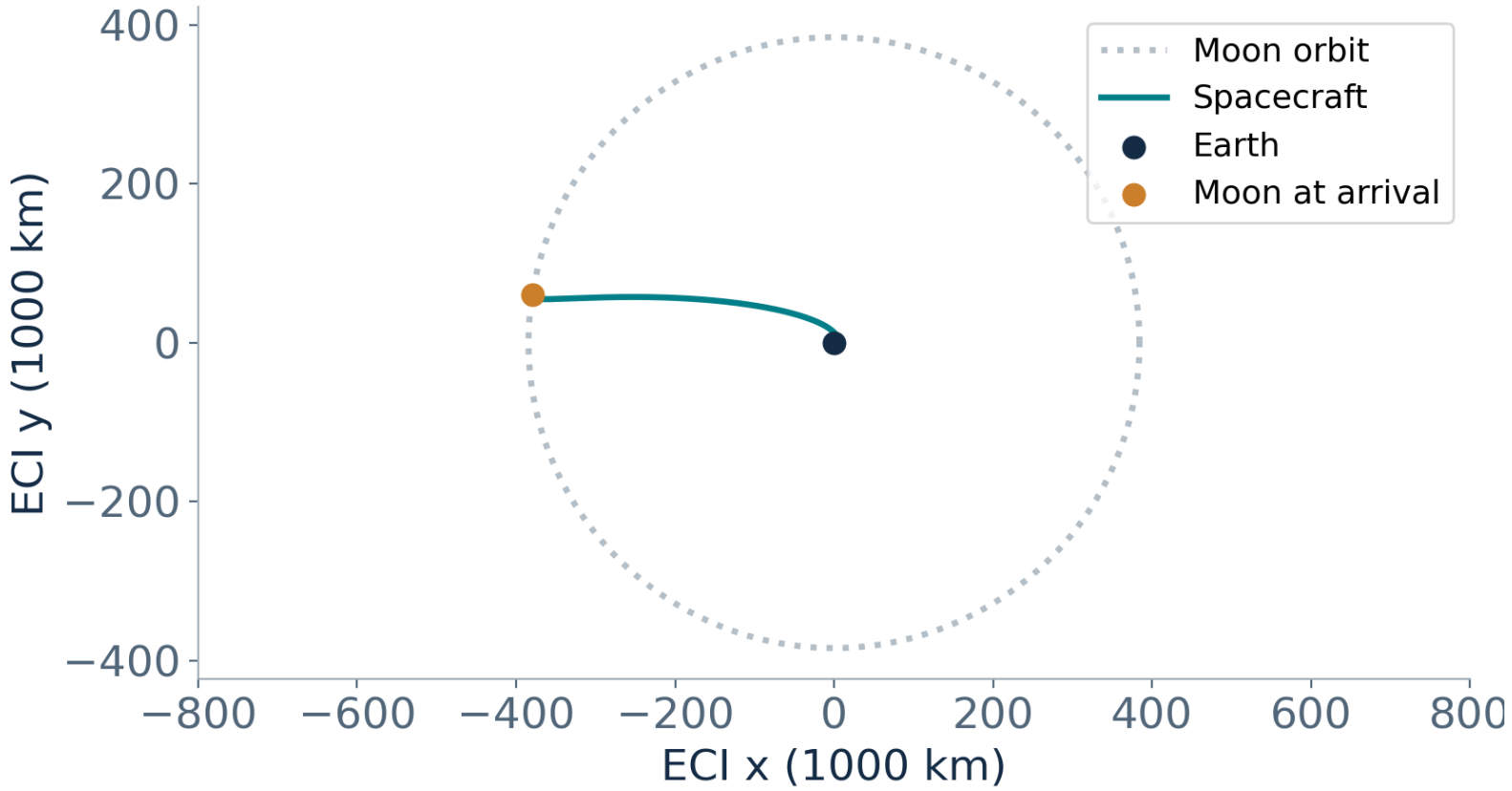
Earth-to-Moon trajectory and lunar capture

Scenario A: Apollo 11

Tasks 1–8: departure and translunar coast
Task 9: lunar orbit insertion

October 20, 2026

Initial conditions and design targets



185 km

Circular parking orbit

110 km

Target lunar altitude

72 h

Chosen travel time after TLI

TLI = trans-lunar injection. Time zero is immediately after the departure impulse.

Physics and the accelerating Earth origin

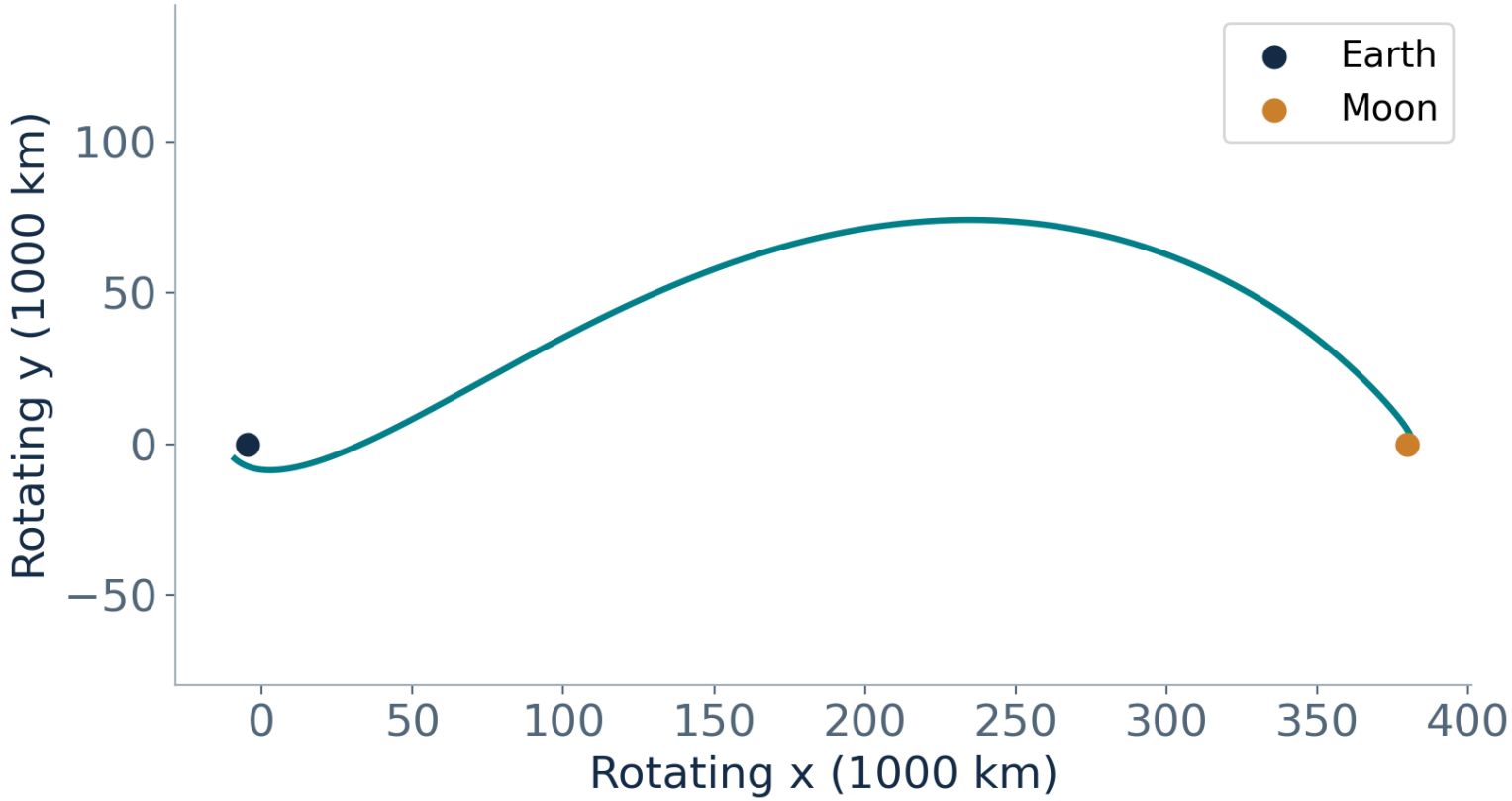
$$\ddot{\mathbf{r}} = -\mu_E \frac{\mathbf{r}}{\|\mathbf{r}\|^3} - \mu_M \left[\frac{\mathbf{r} - \mathbf{r}_M}{\|\mathbf{r} - \mathbf{r}_M\|^3} + \frac{\mathbf{r}_M}{\|\mathbf{r}_M\|^3} \right]$$

r: spacecraft position relative to Earth. r_M: Moon position relative to Earth.
μ_E, μ_M: gravitational parameters. r̈: spacecraft acceleration.

$$m\dot{\mathbf{v}} = \mathbf{F}_g + T\hat{\mathbf{u}}$$

m: spacecraft mass. v: velocity. F_g: gravitational force.
T: thrust. û: thrust unit direction. Exhaust momentum is included in thrust.

Reference frames



Earth-centered

Integrate the flight

Moon-centered

Measure arrival and LOI

Rotating barycentric

Show encounter geometry

LOI = lunar orbit insertion. Full position and velocity transformations: Appendix A.

Own RK4 and the library reference

$$k_1 = f(t, y)$$
$$k_2 = f(t + h/2, y + hk_1/2)$$
$$k_3 = f(t + h/2, y + hk_2/2)$$
$$k_4 = f(t + h, y + hk_3)$$

$$y_{next} = y + \frac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4)$$

10 s

RK4 comparison step

10^{-12}

DOP853 relative tolerance

t: time. h: step. y: state (position, velocity, mass).
f: state derivative. k_1–k_4: stage slopes.

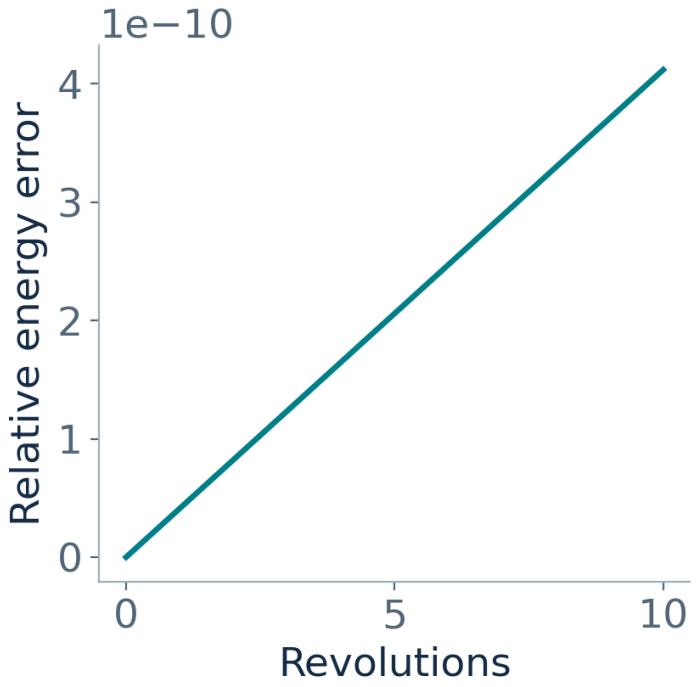
Targeting the lunar encounter

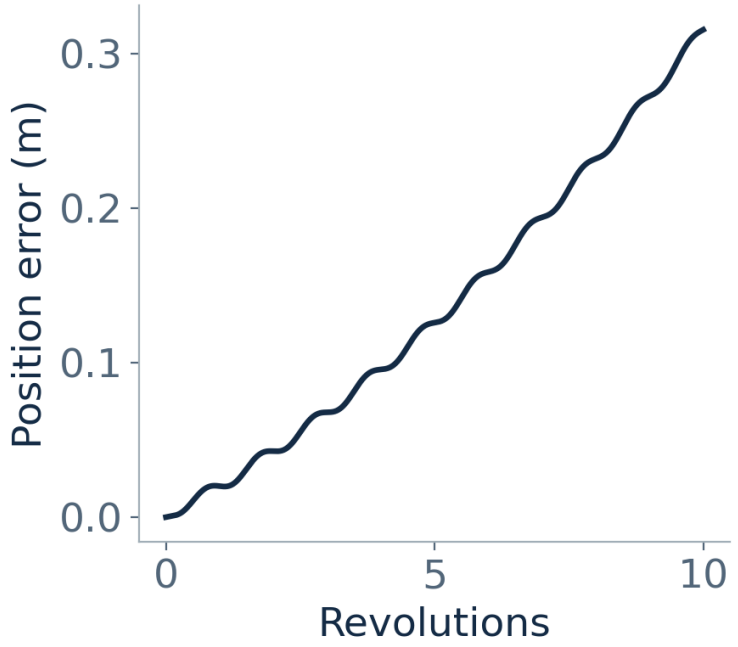
Two unknowns	Two chosen targets
TLI velocity change	110 km lunar altitude
Initial lunar phase	72 h after TLI

$$\boldsymbol{\rho} \cdot \mathbf{v}_{rel} = 0$$

$\rho = r - r_M$: Moon-relative position.
$v_{rel} = v - v_M$: Moon-relative velocity. v_M: Moon velocity.

V1: ten revolutions against an exact orbit





0.315 m

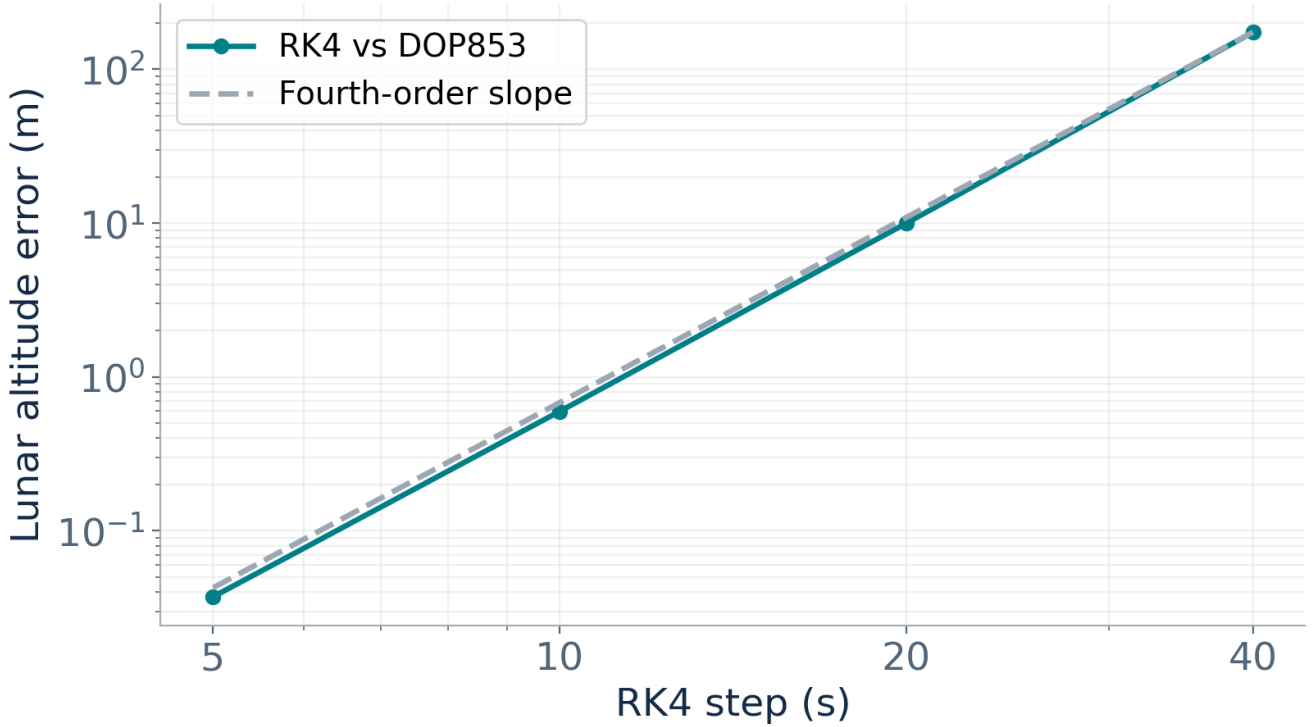
Maximum position error

4.12×10^{-10}

Maximum relative energy error

Earth-only circular motion has an analytic solution. Both error histories cover 10 complete revolutions.

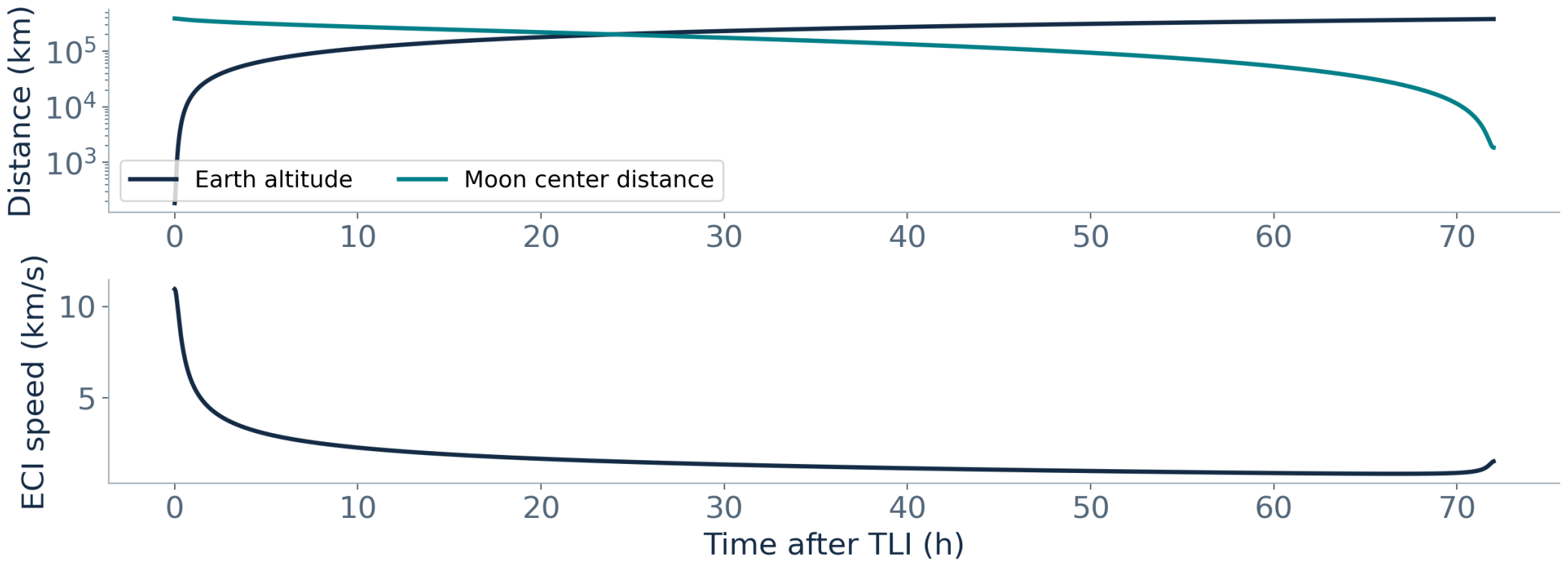
V2 and V3: convergence and agreement



Step (s)	Error (m)
40	174.17
20	9.979
10	0.596
5	0.037

Observed orders: **4.13, 4.06, 4.01**
V3 result: **0.596 m** difference against a **1,000 m** limit.

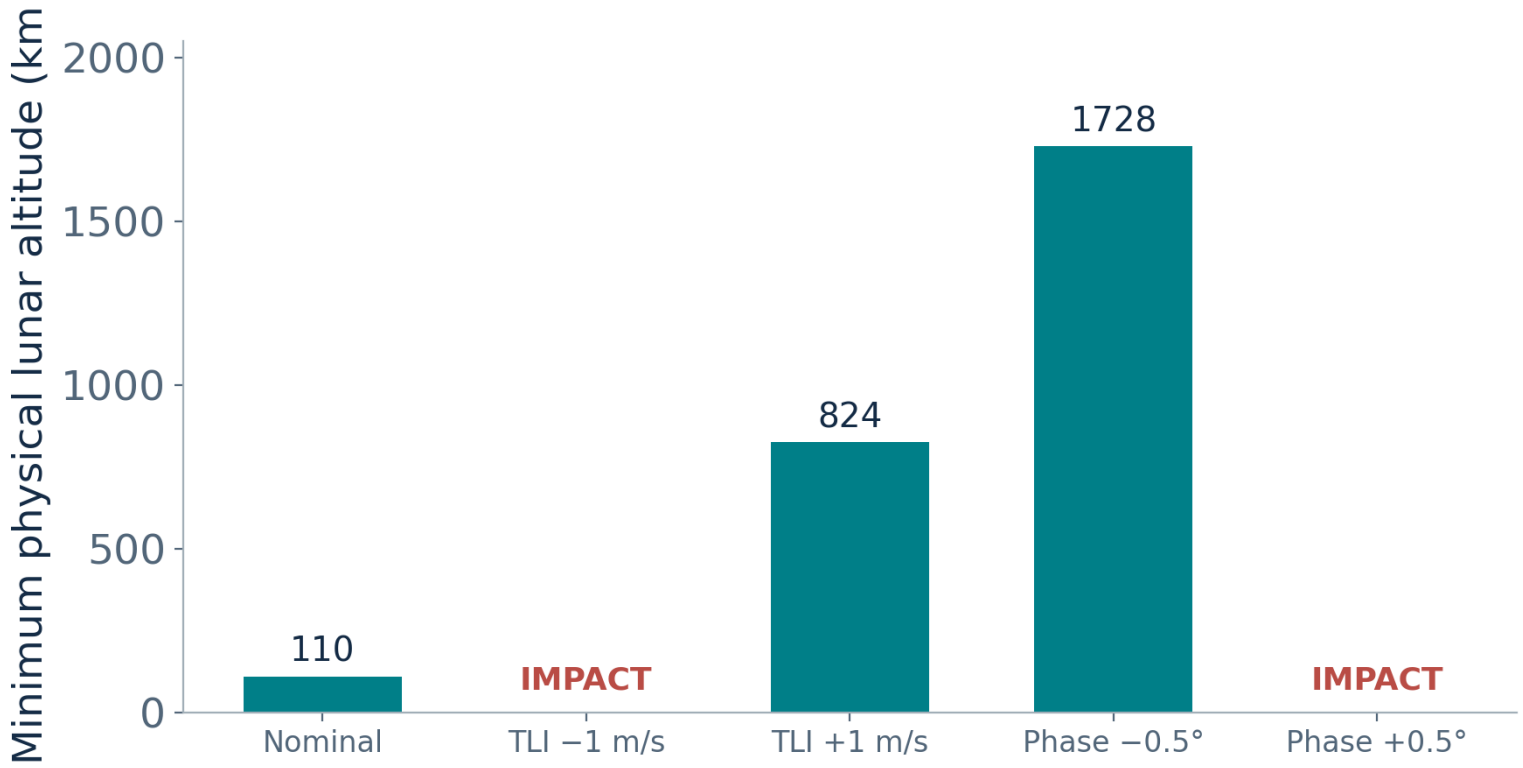
Baseline distance and speed histories



One TLI impulse. Unpowered coast. **110 km lunar altitude at 72.00 h.**

Coast mass = 20,950.9 kg for the stated assumptions. No finite burn is simulated.

Small departure errors change the encounter



Change one input at a time. Keep the other fixed. **No retargeting.**

Impact cases stop at the surface. Their event time is contact time, not a subsurface periapsis.

Baseline summary before lunar capture

185 km

Parking-orbit altitude

3,156.24 m/s

TLI velocity change

72.00 h

Time after TLI

110.00 km

Lunar closest-approach altitude

TLI propellant estimate: **24,049.1 kg** with assumed pre-TLI mass m_0 = 45,000 kg and I_{sp} = 421 s.

Full class-summary table and variable definitions: Appendix C.

Scenario A: the lunar capture impulse

$$v_c = \sqrt{\frac{\mu_M}{r_p}}$$

$$\Delta \mathbf{v}_{LOI} = v_c \hat{\mathbf{t}} - \mathbf{v}_{rel}$$

v_c: circular lunar speed. μ_M: Moon gravitational parameter.
r_p: Moon-center arrival radius. $\hat{t}$: tangent unit vector.
v_{rel}: pre-burn Moon-relative velocity. Δv_{LOI}: capture impulse.

2.529 km/s

Before insertion

1.629 km/s

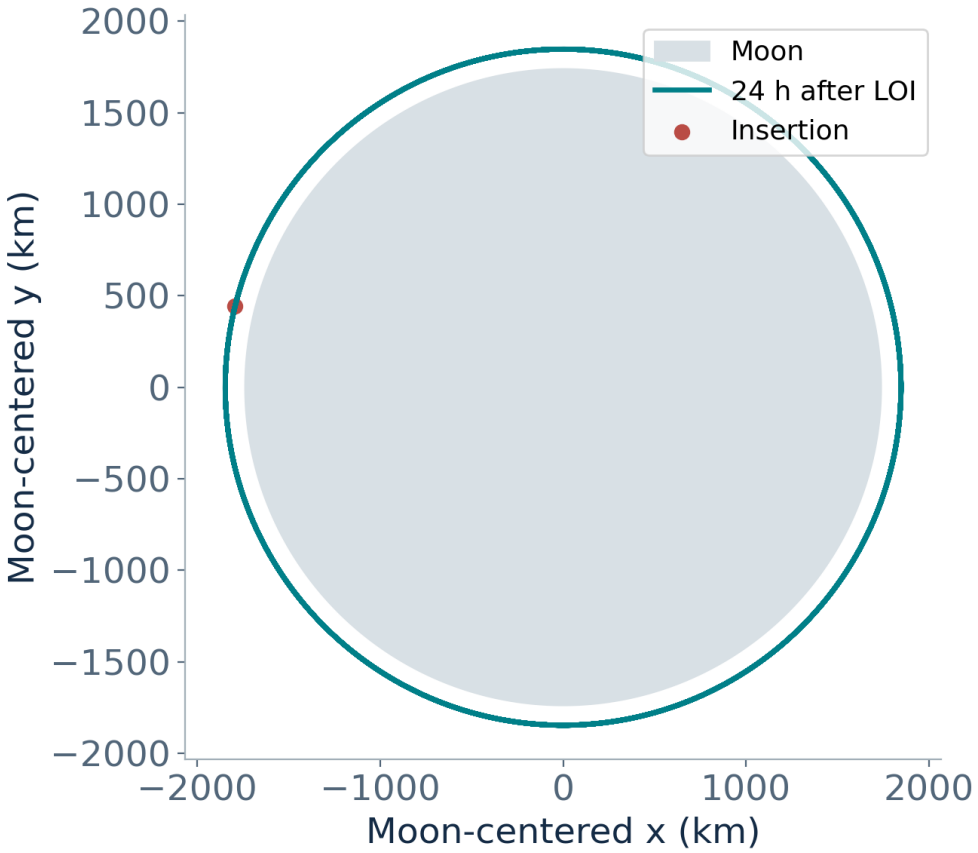
Circular lunar speed

899.80 m/s

LOI velocity change

LOI is an impulse. The burn duration and finite-burn losses are outside this model.

The captured orbit stays near 110 km



109.934–110.011 km

Altitude range over 24 hours

1.979 h

Initial lunar orbital period

An initially circular orbit acquires a small variation under Earth tides.

Comparison with Apollo 11

Maneuver / budget	Velocity change
Model: direct circular insertion	899.80 m/s
Apollo 11: first lunar capture [1]	889.25 m/s
Apollo 11: later circularization [1]	48.40 m/s
Apollo 11: sum of burn magnitudes	937.66 m/s

The real mission used two maneuvers. Our model targets a 110 km circle with one impulse.

[1] NASA, Apollo 11 Mission Report (1969), Table 7-V. Source link: Appendix D.

What the model establishes

A verified translunar coast

A 110 km lunar orbit after a **899.80 m/s** capture impulse

Limits: circular planar lunar motion, instantaneous burns, and illustrative mass / engine assumptions

Reproducible notebook and code. Questions?

Appendix A: exact frame transformations

$$\boldsymbol{\rho} = \mathbf{r} - \mathbf{r}_M, \quad \mathbf{v}_{rel} = \mathbf{v} - \mathbf{v}_M$$

$$\mathbf{q} = R(-\alpha)\mathbf{r} - (\mu a, 0)$$

$$\dot{\mathbf{q}} = R(-\alpha)\mathbf{v} - \mathbf{n} \times [R(-\alpha)\mathbf{r}]$$

$\alpha = nt + \theta_0$. n: Moon mean motion. t: time. θ_0: initial lunar phase.
$\mu = \mu_M / (\mu_E + \mu_M)$: mass ratio. a: Earth–Moon distance.
$R(-\alpha)$: planar rotation with rows $(\cos\alpha, \sin\alpha)$ and $(-\sin\alpha, \cos\alpha)$.
q and $\dot{q}$: barycentric rotating position and velocity. n-vector = $n\,\hat{z}$.
$\hat{z}$: unit normal to the orbital plane.

Appendix B: rocket equation and assumptions

$$T = \dot{m}_p V_e, \quad \dot{m} = -\dot{m}_p, \quad V_e = I_{sp} g_0$$

$$\Delta v = I_{sp} g_0 \ln \frac{m_{before}}{m_{after}}$$

T: thrust. $\dot{m}_p > 0$: outgoing propellant flow. $\dot{m}$: spacecraft mass rate.
V_e: exhaust speed. I_{sp}: specific impulse. $g_0 = 9.80665$ m/s².
Δv: impulse magnitude. m_{before}, m_{after}: masses around an impulse.

Assumed pre-TLI mass: $m_0 = 45{,}000$ kg, TLI $I_{sp} = 421$ s, LOI $I_{sp} = 314$ s.
These estimate propellant. They do not reconstruct historical stage performance.

Appendix C: full class-summary row

Required quantity	Computed value
Parking-orbit altitude	185 km
TLI velocity change	3156.242 m/s
TLI propellant, assumed mass / Isp	24,049.132 kg
Time to lunar closest approach	72.000 h after TLI
Lunar closest-approach altitude	110.000 km
Own integrator / step	RK4 / 10 s
Library reference / tolerance	DOP853 / relative 10^{-12}

LOI propellant is separate: 5,308.698 kg for the stated assumptions.

Appendix D: numerical results and sources

Computed results
MEC568_Tasks1to8_ScenarioA.ipynb
Tasks 1–8 baseline and Task 9 lunar capture. Default 110 km / 72 h targets.

Project requirements and constants
MEC 568 Spacecraft Trajectory Calculation handout, Sections 4–7.

Historical maneuver comparison [1]
NASA, Apollo 11 Mission Report, November 1969, Table 7-V.
2917.5 ft/s and 158.8 ft/s, converted using 1 ft = 0.3048 m.

nasa.gov/wp-content/uploads/static/apollo50th/pdf/A11_MissionReport.pdf

Appendix E: contributions and AI disclosure

Complete actual contributions before submission

Team member	Actual physics, code, verification, and presentation work
Presenter 1	
Presenter 2	
Presenter 3	

AI disclosure: ChatGPT helped implement and verify the simulation and draft the explanations, slides, and presenter notes. Every member must understand and explain the work.